

Highlights

Identifiability of a point defect model for oxide growth on AISI 347

- Point defect model identified for duplex oxide on AISI 347 in reactor water
- Profile likelihood leaves one of five fitted parameters undetermined
- 95% of the fit statistic rests on three points, and 80% on a single one
- Power law and mechanistic fits agree on data but differ 3.5–6.3x at ten years
- An exposure beyond 3000 h would discriminate the two extrapolations